

Project 3

AUTHOR

Savanna Kerby

Project Idea [🔗](#)

For this project, we had complete liberty in what our research questions could be about. With this in mind as well as my exploding grocery expenses, I choose to focus on the last 10 years of the USA Consumer Price Index. The Consumer Price Index (CPI) measures how much the average price of a fixed “basket” of goods and services changes over time. It is often used to track inflation and answers the question: “How much more expensive is life today compared to before”.

My research questions I want to focus on are: What are the current trends with the price of food in the United States in each food group? Is there seasonality trends for the changes in CPI by month? And are things increasingly getting more expensive or are we following the generally recommended 2% inflation increase each year?

My Data Sets

To answer these questions, I found two data sets to merge together. One is from the USDA Economic Research Service Food Price Outlook. It shows us the categorized annual CPIs for specific food groups. The other data set I used was from the U.S. Bureau of Labor Statistics that recorded the monthly CPI for all urban consumers. These data sets differ in that one focused on the food annual CPI while the other records the overall monthly CPI. However, these differences will help us more deeply understand how things currently stand.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.5.2

— Attaching core tidyverse packages — tidyverse 2.0.0 —

```
✓ forcats 1.0.1    ✓ readr 2.1.5
✓ ggplot2 4.0.0    ✓ stringr 1.5.2
✓ lubridate 1.9.4  ✓ tibble 3.3.0
✓ purrr 1.1.0     ✓ tidyr 1.3.1
```

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
# read in data
month_data = read.csv("month_CPI_data.csv")
year_data = read.csv("historicalcpi.csv")

#clean month CPI data
clean_month = month_data %>%
  select(-c(HALF1, HALF2)) %>%
  pivot_longer(
    col = Jan:Dec,
    names_to = "Month",
    values_to = "CPI"
  ) %>%
  mutate(
    mom_pct_change = (CPI - lag(CPI)) / lag(CPI) * 100,
    Month = factor(Month, levels = month.abb)
  )

#get the year over year percent change in CPI
clean_CPI_1 = clean_month %>%
  group_by(Year) %>%
  filter(
    (Year != 2025 & Month == "Dec") |
    (Year == 2025 & Month == "Sep")
  ) %>%
  ungroup() %>%
  arrange(Year) %>%
  mutate(
    yoy_pct_change = (CPI - lag(CPI)) / lag(CPI) * 100
  ) %>%
  select(Year, yoy_pct_change)

#clean year CPI data
clean_year = year_data %>%
  filter(Year > 2014) %>%
  pivot_wider(
    names_from = Consumer.Price.Index.item,
    values_from = Percent.change
  )
```

```
#merge data sets together
CPI_df = clean_month %>%
  left_join(clean_CPI_1, by = "Year") %>%
  left_join(clean_year, by = "Year") %>%
  slice(1:(n() - 3)) %>%
  pivot_longer (
    col = 'All food' : 'Other foods',
    names_to = "food_type",
    values_to = "percent_change"
  ) %>%
  mutate(
    food_type = as.factor(food_type),
    Month = factor(Month, levels = month.abb)
  )

# write.csv(CPI_df, "CPI_cleaned_df.csv", row.names = FALSE)
```

Summary Table

Another objective of this project is to seamlessly use AI combined with my own efforts to create useful graphs and visuals. Since many of the food heading names were long and hard to look at, I will use kable and kableExtra to create a more visually appealing data set using the first 10 lines for this example with help from AI.

AI-SOURCE: ChatGPT (2025-12-10).

Prompt summary: Asked how to create a styled table showing 10 rows using kableExtra/gt.

Your edits: Replaced colors, changed caption wording, selected 10 specific rows.

#AI used to style the table (ChatGPT, 2025-12-10); major edits: adjusted colors and simplified layout

```
library(knitr)
```

Warning: package 'knitr' was built under R version 4.5.2

```
library(kableExtra)
```

Warning: package 'kableExtra' was built under R version 4.5.2

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

```
group_rows
```

```
#only show 10 entries
first_ten = CPI_df %>%
  slice(1:10)

#style the data using AI
first_ten %>%
  kable(
    format = "html",
    caption = "Summary of Cleaned & Transformed Data (First 10 Rows)",
    digits = 2
  ) %>%
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE,
    position = "center"
  ) %>%
  column_spec(1, bold = TRUE) %>% # bold first column
  row_spec(0, bold = TRUE, color = "white", background = "#3b82f6") %>% # header
  footnote(
    general = "Generated with AI styling assistance; manually reviewed for accuracy.",
    footnote_as_chunk = TRUE
  )
```

Summary of Cleaned & Transformed Data (First 10 Rows)

	Year	Month	CPI	mom_pct_change	yoy_pct_change	food_type	percent_change
	2015	Jan	233.71	NA	NA	All food	1.9
	2015	Jan	233.71	NA	NA	Food away from home	2.9
	2015	Jan	233.71	NA	NA	Food at home	1.2
	2015	Jan	233.71	NA	NA	Meats, poultry, and fish	1.9
	2015	Jan	233.71	NA	NA	Meats	3.0
	2015	Jan	233.71	NA	NA	Beef and veal	7.2
	2015	Jan	233.71	NA	NA	Pork	-3.9
	2015	Jan	233.71	NA	NA	Other meats	4.1
	2015	Jan	233.71	NA	NA	Poultry	0.4
	2015	Jan	233.71	NA	NA	Fish and seafood	-0.9

Note: Generated with AI styling assistance; manually reviewed for accuracy.

Figure A - AI Assisted

AI-SOURCE: ChatGPT (2025-12-10).

Prompt summary: Asked AI to generate two side-by-side ggplot figures showing seasonal monthly trends and yearly YOY percent change.

Your edits: Adjusted titles, labels, and theme settings as well as adding a continuous x scale and tilt.

```
library(patchwork)
```

Warning: package 'patchwork' was built under R version 4.5.2

```
#monthly seasonal pattern
p1 <- CPI_df %>%
  ggplot(aes(x = Month, y = mom_pct_change, group = Year, color = as.factor(Year))) +
  geom_line(linewidth = 0.7) +
  scale_color_viridis_d() +
  labs(
    title = "Monthly Seasonal CPI Trends",
    x = "Month",
    y = "CPI",
    color = "Year"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )

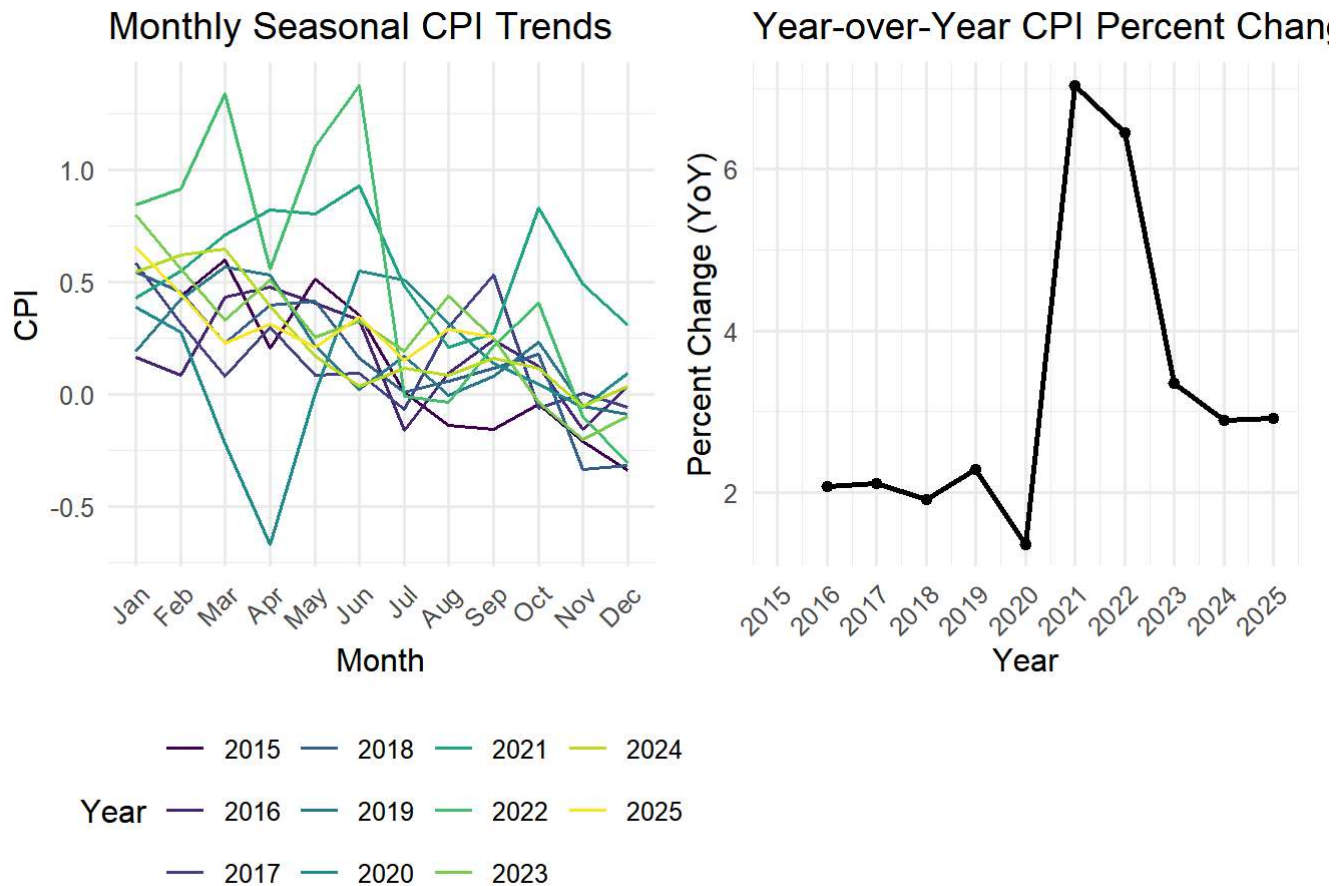
#year-over-year percent change
p2 <- CPI_df %>%
  ggplot(aes(x = Year, y = yoy_pct_change)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(
    title = "Year-over-Year CPI Percent Change",
    x = "Year",
    y = "Percent Change (YoY)"
  ) +
  theme_minimal(base_size = 12) +
  scale_x_continuous(breaks = unique(CPI_df$Year)) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

p1 + p2
```

Warning: Removed 22 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 264 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 264 rows containing missing values or values outside the scale range (``geom_point()``).



From these two graphs, it helps answer two of our research questions. The first was: Is there seasonality trends for the changes in CPI by month? Although the data isn't very consistent, we can see that the graph tends to have a downward trend as the end of the year approaches implying that costs tend to be higher at the beginning of each year. In our other graph, it answers the question: Are things increasingly getting more expensive or are we following the government generally recommended 2% inflation increase each year? From the second graph, we see that before 2020, the CPI was stable around 2%. However, after 2020, it skyrocketed to over 6% and has now stabilized closer to 3%. So although the CPI is above the recommended 2%, it does seem to be slowly returning to a more stable level.

Figure B - No AI

We will now seek to answer our last research question asked above by using my own knowledge and skills to create a graphic.

No AI assistance was used for this figure

```
CPI_df %>%  
  ggplot(aes(x = Year, y = percent_change, color = food_type), group = food_type) +  
  geom_line() +  
  geom_point() +  
  theme_minimal() +  
  scale_x_continuous(breaks = unique(CPI_df$Year)) +  
  theme(  
    axis.text.x = element_text(angle = 45, hjust = 1),  
    legend.text = element_text(size = 8),  
    legend.title = element_text(size = 9),  
    legend.key.size = unit(0.4, "cm")  
  ) +  
  labs(  
    x = "Year",  
    y = "CPI percent change",  
    title = "Year CPI percent change for different food groups in the US"  
  )
```

Warning in fortify(data, ...): Arguments in `...` must be used.

✖ Problematic argument:

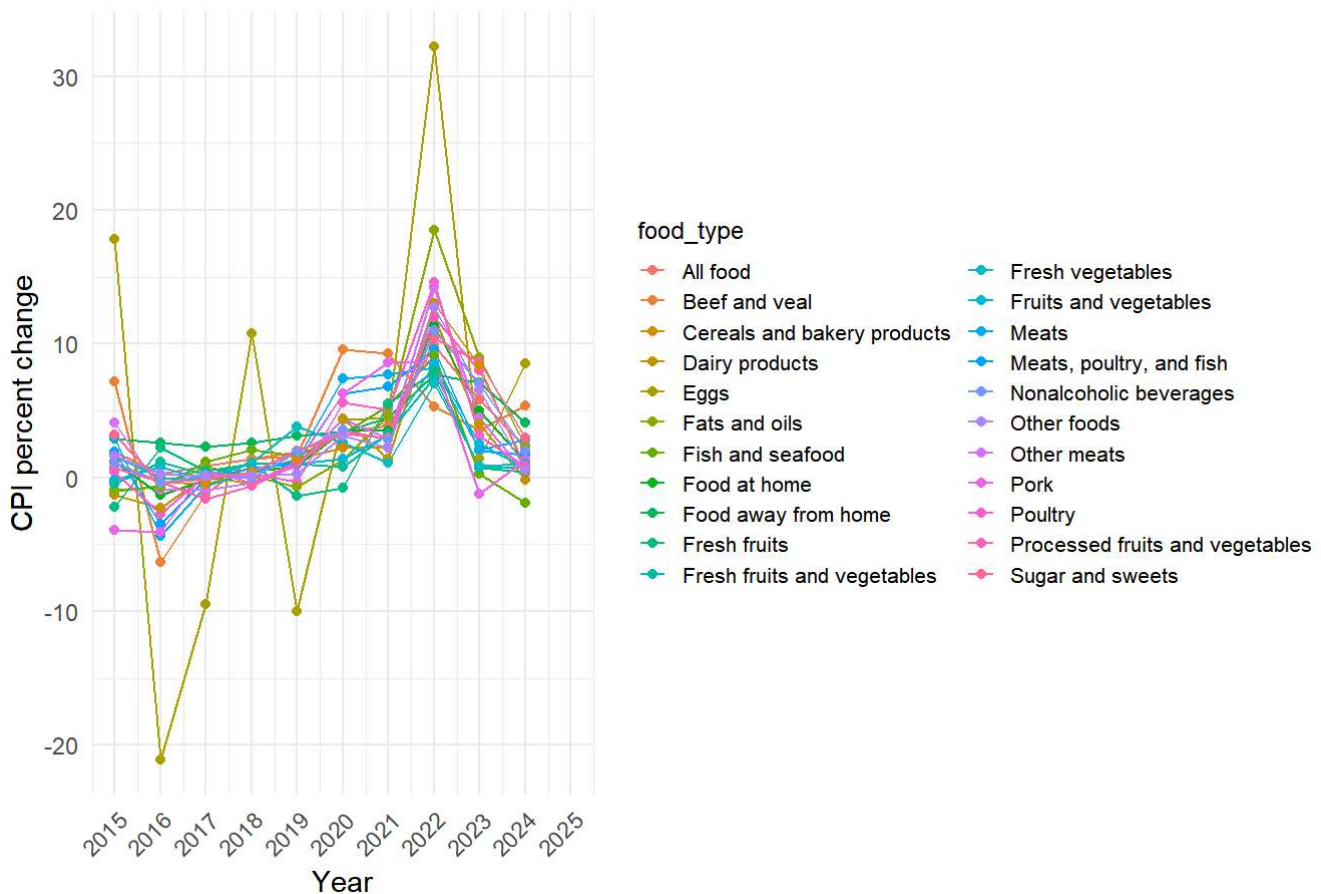
- group = food_type

ℹ Did you misspell an argument name?

Warning: Removed 198 rows containing missing values or values outside the scale range (`geom\_line()`).

Warning: Removed 198 rows containing missing values or values outside the scale range (`geom\_point()`).

Year CPI percent change for different food groups in the US



Our last question was: What are the current trends with the price of food in the United States in each food group? From this graphic, we see that in the past 10 years that eggs have definitely been the most variable food item and that although there was a spike in cost of food in 2022, that for the most part, these prices have gone back down to a more reasonable level.

Reusable Theme/Helper - AI Contribution

I choose to do a reusable theme/helper for my additional AI contribution as I feel like this is something that would be quite useful as I used much of this code repeatably. I'd also help me not forget to do more basic things that enhance the look of my graphics. I'll apply this theme back onto my "Monthly Seasonal Trends" graph so you can see what it does.

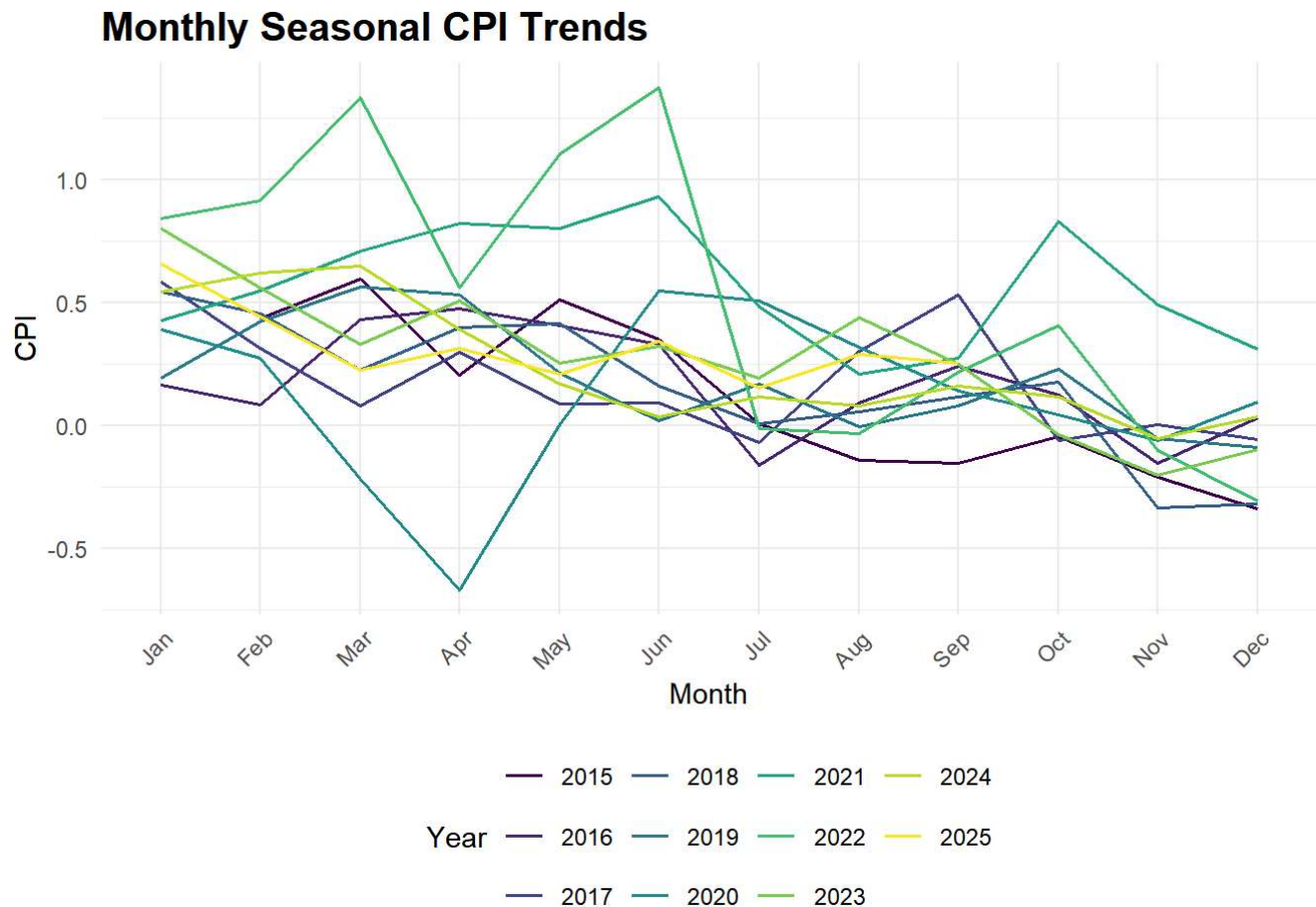
AI-SOURCE: ChatGPT (2025-12-10)

Prompt summary: create a simple custom ggplot theme; Your edits: tweaked font sizes.

```
theme_cpi = function() {
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom",
    plot.title = element_text(face = "bold", size = 15)
```

```
)  
}  
  
#apply it  
p1 + theme_cpi()
```

Warning: Removed 22 rows containing missing values or values outside the scale range (`geom\_line()`).



API Appendix

Key AI Prompts Used

- “How do I make a theme function for ggplot?” → AI drafted the initial version of `theme_cpi()`, and I customized spacing, font size, and colors.
- “How do I merge my year and month CPI datasets correctly?” → AI helped debug `left_join()` and spotting grouping/lag issues.
- “Why is my year-over-year percent change calculating incorrectly?” → AI explained lagging by year end and creating a special case for 2025.

Before/After Code Snippet

AI Output (before editing)

```
#monthly seasonal pattern
p1 <- CPI_df %>%
  ggplot(aes(x = Month, y = month_change, group = Year, color = as.factor(Year))) +
  geom_line(linewidth = 0.7) +
  scale_color_viridis_d() +
  labs(
    title = "Monthly Seasonal CPI Trends",
    x = "Month",
    y = "CPI",
    color = "Year"
  ) +
  theme_minimal(base_size = 12)

#year-over-year percent change
p2 <- CPI_df %>%
  ggplot(aes(x = Year, y = yoy_pct_change)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(
    title = "Year-over-Year CPI Percent Change",
    x = "Year",
    y = "Percent Change (YoY)"
  ) +
  theme_minimal(base_size = 12)

p1 + p2
```

My Edited Version

```
#monthly seasonal pattern
p1 <- CPI_df %>%
  ggplot(aes(x = Month, y = mom_pct_change, group = Year, color = as.factor(Year))) +
  geom_line(linewidth = 0.7) +
  scale_color_viridis_d() +
  labs(
    title = "Monthly Seasonal CPI Trends",
    x = "Month",
    y = "CPI",
    color = "Year"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
```

```
#year-over-year percent change
p2 <- CPI_df %>%
  ggplot(aes(x = Year, y = yoy_pct_change)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(
    title = "Year-over-Year CPI Percent Change",
    x = "Year",
    y = "Percent Change (YoY)"
  ) +
  theme_minimal(base_size = 12) +
  scale_x_continuous(breaks = unique(CPI_df$Year)) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

p1 + p2
```

Key edits:

- Added a continuous x scale for the Year
- Angled the x-axis text to fit better
- Corrected the aes() to the correct variable names

Reflection

AI helped me confidently write my code where my experience lacked at times. I used it to debug areas I got stuck on, like when it helped me realize I needed to add another pivot longer in order to graph by food_types later in my project. This led to my data being even more tidy, something I hadn't realized was missing before. I made sure to test all generated code and edited it to match my project structure, preferences, and data set names. I only used things I understood and took time to learn helpful things it suggested like how to use patchwork. I verified the correctness with each intermediate step by using the View() function to inspect my data frame and the values being used to generate the graphs. In the future, I'd want to use AI more just for debugging and use more of my own knowledge to code functions or to do the necessary research to use functions I haven't in the past. Overall, AI was useful for troubleshooting and refining my visualizations while I maintained full control over the analysis and direction of my report.